

function *nama_fungsi* (param1: type1, param2: type2) -> type_hasil

{Spesifikasi fungsi}

KAMUS LOKAL

ALGORITMA

-> hasil

procedure *nama_proc*(input/output param1: type1, ...)

{I.S. : ; F.S. :}

KAMUS LOKAL

ALGORITMA

Studi Kasus

Program.....

KAMUS

M : array[1..3] of array[1..4] of integer

procedure InputMatriks (output M : array[1..3] of array[1..4] of integer)

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i, j : integer

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 i traversal[1..4]

 j traversal[1..3]

input(M[i][j])

procedure Operasi2Matriks (input M1 : array[1..3] of array[1..4] of integer; input M2 : array[1..3] of array[1..4] of integer, input op : character; output MHasil : array[1..3] of array[1..4] of integer)

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i, j : integer

ALGORITMA

 i traversal[1..4]

 j traversal[1..4]

if(op = "+") then

 MHasil[i][j] <- M1[i][j]+M2[i][j]

else if(op = "-") then

 MHasil[i][j] <- M1[i][j]-M2[i][j]

procedure CetakMatriks(input M : array[1..3] of array[1..4] of integer)

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i, j : integer

ALGORITMA

```
i traversal[1..4]
  j traversal[1..4]
    output(M[i][j], " ")
```

Bisa pake ADT :)))))))))

Latihan Soal 1

```
function Max2 (a : real, b: real) -> real
  if(a > b) then
    max <- a -> a
  else
    max <- b -> b
  -> max ( ini gausah lagi klo udh ikut yg merah
```

```
function Max3 (a: real, b: real, c: real) -> real
```

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```
temp : integer
```

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```
temp <- Max2(a, b)
max <- Max2(c, temp)
-> max
```

Latihan Soal 2

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```
type Tanggal : < DD : integer[1...31];
              MM : integer[1...12];
              YY : integer > 0 >
```

T : Tanggal

```
function LeapYear(T : Tanggal, cek : integer) -> integer
```

```
  if(T.YY mod 4 = 0 and not(T.YY mod 100 = 0) and not(T.YY mod 400 = 0)) then
    cek <- 1
  -> cek
```

```
function NextDay(T : Tanggal)->Tanggal
```

```
  depend on (T)
    LeapYear(T.YY) = 1 and T.MM = 2 :
      if(T.DD = 29) then
        T.DD <- 1
        T.MM <- 3
      else
        T.DD <- T.DD + 1
```

T.DD = 31 and T.MM = 12 :

T.DD <- 1

T.MM <- 1

T.YY <- T.YY + 1

T.DD = 30 and (T.MM = 2 or T.MM = 4 or T.MM = 6 or T.MM = 9 or T.MM = 11):

T.DD <- 1

T.MM <- 1

T.DD = 31 and (T.MM = 1 or T.MM = 3 or T.MM = 5 or T.MM = 7 or T.MM = 8 or
T.MM = 10) :

T.DD <- 1

T.MM <- 1

else

T.DD <- T.DD + 1

Latihan Soal 3

procedure Tukar(input/output a : integer; input/output b : integer)

temp <- a

a <- b

b <- temp

procedure Putar3Bil (input/output a : integer; input/output b : integer; input/output c : integer)

Tukar(a, c)

Tukar(b, c)

Latihan Soal 4

KAMUS

type Tanggal : < DD : integer[1...31];

MM : integer[1...12];

MM2 : string;

YY : integer > 0 >

function LeapYear(T : Tanggal, cek : integer) -> integer

cek <- 0

if(T.YY mod 4 = 0 and not(T.YY mod 100 = 0) or (T.YY mod 400 = 0)) then

cek <- 1

-> cek

function NamaBulan(T : Tanggal) -> Tanggal

depend on (T)

T.MM = 1 :

T.MM2 <- "Januari"

```

T.MM = 2 :
    T.MM2 <- "Februari"
T.MM = 3 :
    T.MM2 <- "Maret"

T.MM = 4 :
    T.MM2 <- "April"
T.MM = 5 :
    T.MM2 <- "Mei"
T.MM = 6 :
    T.MM2 <- "Juni"
T.MM = 7 :
    T.MM2 <- "Juli"
T.MM = 8
    T.MM2 <- "Agustus"
T.MM = 9 :
    T.MM2 <- "September"
T.MM = 10 :
    T.MM2 <- "Oktober"
T.MM = 11 :
    T.MM2 <- "November"
T.MM = 12 :
    T.MM2 <- "Desember"

```

function IsTanggalValid(d : integer, m : integer, y : integer) -> integer

```

cek <- 0
if((T.DD <= 31 and (T.MM = 1 or T.MM = 3 or T.MM = 5 or T.MM = 7 or T.MM = 8 or T.MM
= 10 or T.MM = 12)) or (T.DD <= 30 and (T.MM = 2 or T.MM = 4 or T.MM = 6 or T.MM = 9
or T.MM = 11)) or (T.MM = 2 and ((LeapYear = 1 and T.DD <= 29) or (LeapYear = 0 and
T.DD <= 28)) then
    cek <- 1
-> cek

```

procedure BacaTanggal(input d: integer; input m: integer; input y: integer; output T : Tanggal) -> Tanggal

```

if(IsTanggalValid(d, m, y) = 0) then
    while((d > 31 and (m = 1 or m = 3 or m = 5 or m = 7 or m = 8 or m = 10 or m =
12)) or (d > 30 and (m = 2 or m = 4 or m = 6 or m = 9 or m = 11)) or (m = 2 and
((LeapYear = 1 and d > 29) or (LeapYear = 0 and d > 28)) then
        output("Tanggal tidak valid")
        input(T.DD, T.MM, T.YY)
T.DD <- d
T.MM <- m
T.YY <- y

```

function NextDay(T : Tanggal) -> Tanggal

depend on(T)

LeapYear(T.YY) = 1 and T.MM = 2:

if(T.DD = 29) then

T.DD <- 1

T.MM <- 3

else

T.DD <- T.DD + 1

T.DD = 31 and T.MM = 12 :

T.DD <- 1

T.MM <- 1

T.YY <- T.YY + 1

T.DD = 30 and (T.MM = 2 or T.MM = 4 or T.MM = 6 or T.MM = 9 or T.MM = 11));

T.DD <- 1

T.MM <- 1

T.DD = 31 and (T.MM = 1 or T.MM = 3 or T.MM = 5 or T.MM = 7 or T.MM = 8 or
T.MM = 10));

T.DD <- 1

T.MM <- 1

else

T.DD <- T.DD + 1

NamaBulan(T)

procedure TulisTanggal(input T : Tanggal)

output(T.DD, T.MM2, T.YY)